

# Quantum GNN for Supplier Procurement Risk Analysis

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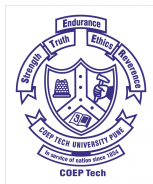
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Presented by  
Aditya Jagtap  
712522002

Project Guide  
Dr. Amit D. Joshi



Department of Computer Science and Engineering  
COEP Technological University (COEP Tech), Pune  
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**Name:** Aditya Jagtap

**MIS:** 712522002

**Degree:** M.Tech (2025–2027)

**Branch:** Computer Engineering

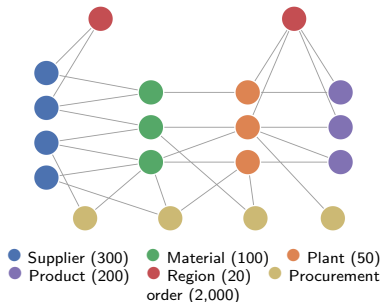
**Department:** Computer Science and Engineering,  
COEP Technological University Pune

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# Introduction

- **Procurement:** sourcing materials from external suppliers so production can continue
- A supply chain is a **network**; a problem at one node **propagates along it**
- **Risk assessment:** predict which suppliers are likely to be disrupted, and how severely
- Hard: disruptions are **rare** (class imbalance) and **severe** events differ from normal history



Illustrative subgraph; full graph: 2,670 nodes, 7,675 edges

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- **GNN:** each node builds an embedding by aggregating its neighbours' information. **GraphSAGE** does this by sampling neighbours, so it scales
- For risk: suppliers are nodes, dependencies are edges, output is a **risk score per supplier**
- Classical limits: **over-smoothing** and **over-squashing**
- **QGNN:** a variational quantum circuit replaces part of the model. Thought to help (hypotheses we test):
  - Encoding into a **high-dimensional Hilbert space**
  - **Entanglement** models feature correlations compactly
  - Very few parameters, suited to small data and rare events
- **Our design:** a small VQC head on a **frozen GraphSAGE embedding**, as a controlled comparison with classical GraphSAGE — not a quantum-advantage claim

# Literature Review

Table 1: Literature Survey

| Ref | Author (Year)             | Focus / Method                                                                        | Limitation                                                         |
|-----|---------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| [1] | Fareed (2026)             | Classical ML for supplier risk prediction and resilience scoring                      | Preprint, not peer-reviewed; purely classical, no quantum baseline |
| [2] | Ziaria et al. (2026)      | ML-based supplier evaluation; sustainable supplier selection and delay-risk framework | Classical ML only; no graph structure or quantum component         |
| [3] | Rahman & Nguyen (2026)    | QCNN survey; entanglement-based correlation learning                                  | Survey only; no supply-chain or graph evaluation                   |
| [4] | Lamichhane & Rawat (2025) | Survey of QML: advances, challenges, perspectives                                     | Survey only; no empirical evaluation on any domain task            |
| [5] | Larbi et al. (2025)       | Dual ML models for IoT component failure and parts demand                             | Classical ML only; no quantum formulation                          |

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# Literature Survey

Table 1: Literature Survey (Contd.)

| Ref  | Author (Year)            | Focus / Method                                                                     | Limitation                                                                            |
|------|--------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| [6]  | Innan et al. (2024)      | QGNN with VQC for credit-card fraud detection                                      | Simulator only; 6 qubits; heavy undersampling; $\approx 3\%$ gain over classical GNN  |
| [7]  | Tudisco et al. (2024)    | VQC classifier on real bank transaction data                                       | Dataset reduced in size and features via PCA due to NISQ limits; scalability unproven |
| [8]  | Soni et al. (2024)       | Survey of quantum optimization in supply chain management and logistics            | Survey only; no learned risk-prediction model                                         |
| [9]  | Jiang, Shen & Liu (2022) | Quantum inventory model; supply-chain structures suited to quantum methods         | Inventory setting; no learned risk prediction or GNN                                  |
| [10] | Yadav (2021)             | Survey of GCN/GAT risk propagation; GNNs model multi-hop supply-chain dependencies | Survey only; classical GNNs, no quantum formulation                                   |

# Motivation and Need for Study

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- **Same structural problem as fraud:** rare, high-impact events in a relational network with extreme class imbalance — the setting where QGNN and VQC work has been reported [6, 7]
- **Supplier networks are natively graphs:** suppliers, hubs and raw-material sources as nodes; dependency and material flow as edges
- **Classical GNNs struggle where procurement risk lives:** over-smoothing flattens embeddings at depth, over-squashing bottlenecks multi-hop signal
- **Gap in the surveyed work:**
  - Supplier-risk studies are purely classical [1, 2, 5]
  - Quantum studies target finance, are surveys, or address inventory [3, 4, 6–9]
  - No QGNN applied to supplier procurement risk
- **Need:** a **controlled comparison** against a classical GraphSAGE baseline, so the outcome is defensible either way

## Problem statement

Comparative Study of Supplier Procurement Risk Analysis using GNN in Classical and Quantum Computing Environments

To investigate whether a **hybrid quantum-classical head**, a small variational quantum circuit reading a frozen GraphSAGE supplier embedding, can match or exceed a **classical GraphSAGE** baseline for **supplier disruption-risk prediction**, and to identify which architectural and training factors govern its behavior.

- 1 **Formulate** supplier procurement risk as a node-level prediction task on a supply-chain graph
- 2 **Implement** a classical GraphSAGE baseline for supplier disruption-risk prediction
- 3 **Implement** a hybrid QGNN: a variational quantum circuit head on the frozen GraphSAGE embedding
- 4 **Compare** the classical and quantum models under identical data, splits and metrics
- 5 **Analyze** which design factors (encoding, qubits, depth) affect the hybrid model's behavior

# Software Requirements Specification

**Table 2: Functional and Non-Functional Requirements**

| Requirement                           | Description                                                                                                       |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <i>Functional</i>                     |                                                                                                                   |
| Data ingestion and graph construction | Load the synthetic supplier dataset and build a heterogeneous supply-chain graph (6 node types, 9 relation types) |
| Feature encoding                      | Encode static and time-varying supplier attributes as node features                                               |
| Hybrid risk model                     | GraphSAGE encoder with a VQC head producing a supplier disruption-risk score                                      |
| Disruption simulation                 | Black-swan events simulated over 104 periods (about 2 years, weekly)                                              |
| Evaluation and reporting              | PR-AUC and related metrics on temporal and severity splits, with tables and figures                               |
| <i>Non-functional</i>                 |                                                                                                                   |
| Fair comparison                       | Classical and quantum models share the same data, splits and metrics                                              |
| Reproducibility                       | Fixed random seeds and saved experiment artifacts                                                                 |
| Scalability                           | Pipeline designed for larger graphs (currently 2,670 nodes, 7,675 edges)                                          |

# SRS Contd...: Technology Stack

**Table 3: Technology Stack**

| Category                 | Technology / Tool                                                                        |
|--------------------------|------------------------------------------------------------------------------------------|
| Programming language     | Python 3.12                                                                              |
| Deep learning            | PyTorch                                                                                  |
| Graph learning           | PyTorch Geometric (HeteroConv, SAGEConv)                                                 |
| Quantum framework        | PennyLane (default.qubit, ideal noise-free simulator)                                    |
| Graph analysis           | NetworkX (topology validation)                                                           |
| Classical ML diagnostics | scikit-learn                                                                             |
| Dataset                  | Synthetic multi-tier supply-chain graph benchmark: 2,670 nodes, 7,675 edges, 104 periods |
| Hardware                 | CPU only; no physical quantum hardware used                                              |

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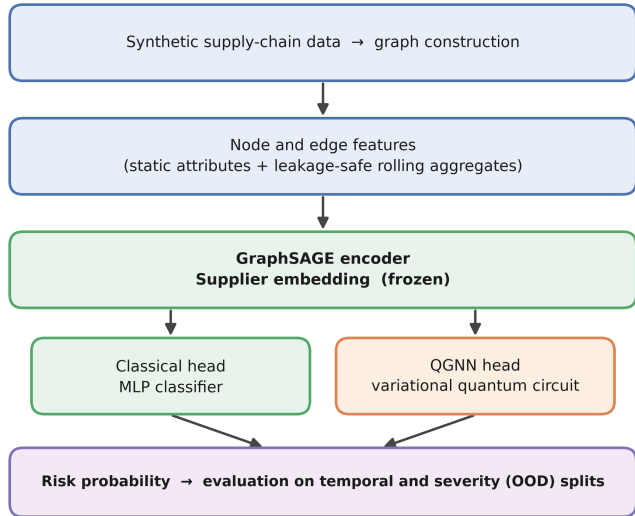
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# SRS Contd...: System Diagram



# Concrete Future Plan

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- **Benchmark and graph:** Synthetic supply-chain dataset with black-swan events; heterogeneous graph construction
- **Classical baseline:** GraphSAGE encoder trained and evaluated on temporal and severity splits
- **QGNN baseline:** VQC head on the frozen GraphSAGE embedding; training stabilisation (patience, output scaling, LayerNorm)
- **Final evaluation:** Classical vs. QGNN over multiple seeds: ranking, calibration and stability metrics

# Expected Outcomes

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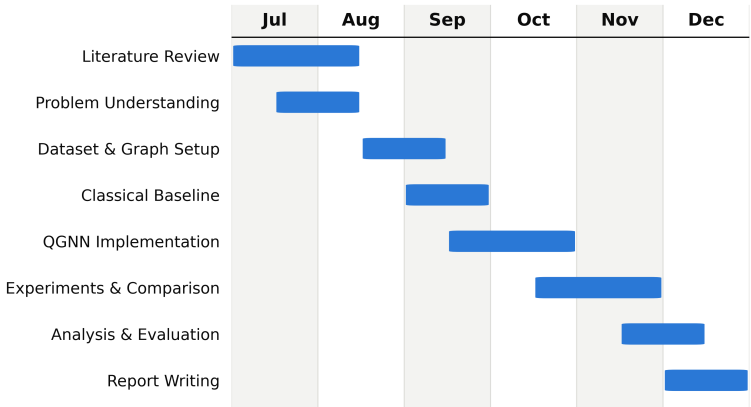
- **Working classical GraphSAGE baseline and hybrid QGNN** for supplier disruption-risk prediction — both implemented on one shared data and training pipeline
- **A controlled comparison of the two models** — same data, splits, metrics and multiple seeds; PR-AUC as primary metric
- **Understanding of normal vs. severe-condition behavior** — temporal split and severity (out-of-distribution) split, reported separately
- **Clearly scoped conclusions** — stated for a synthetic benchmark on an ideal simulator, not as a general quantum-advantage claim

# Summary

- **Problem:** supplier procurement risk is a rare-event prediction task on a network where disruptions propagate
- **Gap:** supplier-risk studies are classical, and quantum studies target other domains such as finance
- **Idea:** pair a classical GraphSAGE encoder with a small variational quantum circuit head for supplier risk prediction
- **Approach:** a controlled comparison against a classical GraphSAGE baseline, studying which design factors matter
- **Expected contribution:** an evidence-based, clearly scoped understanding of when a quantum head helps, not a claim of quantum advantage

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# Timeline Chart Contd...

**Table 6: Project Timeline (July – December)**

| Month             | Planned Work                                                                                                                                                                         |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>July – Aug</b> | Literature review and problem understanding; defining the supplier procurement-risk problem and reviewing recent GNN/QGNN literature                                                 |
| <b>September</b>  | Synthetic supply-chain dataset and heterogeneous graph construction; begin classical GraphSAGE baseline                                                                              |
| <b>October</b>    | Complete classical GraphSAGE baseline; design and implement the QGNN — VQC head on the frozen GraphSAGE embedding, with training stabilisation (patience, output scaling, LayerNorm) |
| <b>November</b>   | Run experiments comparing classical vs. QGNN on temporal and severity splits; begin analysis and evaluation                                                                          |
| <b>December</b>   | Complete analysis and evaluation (ranking, calibration, stability metrics); write the dissertation report                                                                            |

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# Questions/ Discussion